Prototyping A Virtual Reality Game for Commonsense Knowledge Acquisition and Story Understanding

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Abstract

In this paper we introduce our virtual reality game prototype meant for acquiring world knowledge. We present examples of Games With A Purpose (GWAPs) representing different types of media and propose our own by describing its development and initial tests. After describing problems and possible solutions for the project, we discuss how sophisticated gaming could help with collecting more complicated, contextual knowledge from both close to reality and fantasy worlds simulations. With this paper we want to provoke a discussion about the future of virtual reality-based, interactive methods for acquiring common sense needed for better story understanding which is necessary in many domains requiring artificial thinking.

1 Introduction

For ages human thinking processes have been puzzling philosophers and still are a mystery for scholars, including those who pursue the goal of creating intelligent machines. If these machines are to replace human beings in some burdensome domains, they need to be able to know our reasons, our behavior and its consequences. There are many tasks that are dangerous and detrimental to human health, for example mining or operating in s radioactive environment. The current problem of AI is that these tasks require a degree of intelligence machines do not yet possess in order to be completed properly. It is often said that without common sense (meaning also *common* with humans), machines will have problems with "understanding" us [Levesque, 2017] or with keeping us safe [Rzepka and Araki, 2005]. There are various reasons which make commonsense knowledge and reasoning difficult, but it often lies in its commonness. Humans do not usually write down things they deem obvious, and they rarely take pictures of things they see all the time. Common sense is rooted in the way we experience the world, and how to make robots able to experience the world themselves is one of the hardest problems of the field. However, in the new era of Big Data and novel technologies being available to masses, new possibilities arise. Image and video understanding fields are constantly growing and annotators are often asked to identify and name objects [Kazemzadeh *et al.*, 2014]. Annotated images can be used for increasing common sense knowledge, which has been main acquired from text [Singh *et al.*, 2002; Rzepka *et al.*, 2006; Suchanek *et al.*, 2007; Tandon *et al.*, 2014; Krawczyk *et al.*, 2016]. The problem with annotations is that it is costly and/or boring and for that reason GWAPs (Games With A Purpose) approach can become helpful.

2 State of The Art

Acquiring knowledge through games is a relatively new idea but its popularity has been growing steadily. One of the most successful GWAPs, or serious games as they are also called [Marsh, 2011], in this domain was Verbocity [Von Ahn *et al.*, 2006], a game which does not explicitly ask players to add facts as in classic interfaces as OMCS [Singh *et al.*, 2002], where logged in users manually input knowledge, but instead it introduces gamification factor. Users are given one of two roles: Narrator or Guesser. The Narrator uses templates to give hints regarding a word and the Guesser has to guess the word. The use of templates is both element of the game and a way to ease the classification of facts. Fixed templates also make it easier to play against a bot. Other examples of such simple games, meant for extending OMCS, were Common Consensus [Lieberman *et al.*, 2007] and 20 Questions [Speer *et al.*, 2009]. Method used in these games allows to check the validity of the previously acquired facts by replaying them. The knowledge pieces should be independent from each other but keep some relevance. Points are assigned to facts and users increase the score if the fact led to a correct guess of the word and decreases otherwise. Thresholds allow the bot to flag an entry as irrelevant or validated. Verbosity was evaluated by players as both enjoyable and accurate (most of the invalidated facts were debatable statements, such as "Buddhism is a religion").

A different approaches for collecting knowledge for Chinese [Kuo *et al.*, 2009] and Japanese [Nakahara and Yamada, 2011] languages were chosen. The former used an application in which a virtual pet was taught common sense and the latter utilized an image of a little foreign girl who was learning about the world and users were supposed to help her by playing riddles. The virtual pet game utilized an application which already existed in Taiwanese Bulletin Board System PTT. This allowed the developers to simply add game functions rather than designing it from scratch. PPT is accessi-

ble via a text-only terminal interface and has the largest social network in Taiwan with over 800,000 log-ins per day on average [Kuo *et al.*, 2009]. By playing the "Commonsense Q&A", the virtual pet can gain commonsense points that can be traded for food. In this phase of the game, players can either ask their pets a question or answer a question from their pets. Once a question is answered, the player who asked the question will be notified of the answer, as if their pet had learned it from the "pet school". Inside the "pet school" scenario, players can evaluate the answers as good or bad, and if an answer has too many bad ratings, the person who gave this answer is notified not to teach bad things to his or her pet. This allows not only to gather the commonsense facts but also collectively validate them. The entertainment and social aspect of both these game proved useful to gather data of good quality (players feel ashamed to give a bad answer publicly in one case, and want to care for their pets in the other).

In case of Nadya, the Japanese site which was built to gather common sense from people who accessed the discontinued project Nadya.jp and decided to play a guessing game with a system without need of logging in. Users were given riddles with missing words and were given three chances of guessing a related word correctly. Even if all words did not match and a player lost, system was gaining new possible related items. Although the project was not as fruitful as other similar projects [Krawczyk *et al.*, 2016], researchers have lately added an interactive mode to Nadya algorithm [Otani *et al.*, 2016]. The system acquires knowledge in Japanese with the help of a quiz game introduced as a module of a smartphone spoken dialogue system Yahoo! Voice Assist[1]. The approach is similar to the text-based riddle system, however the implementation of the game into already popular dialogue system resulted in the acquisition of a large amount of Japanese common sense knowledge. Authors claim the gathered information is intended to be incorporated into the ConceptNet.

Two main problems with such applications are as follows: a) users tend to get bored quickly; b) they start to input funny / original / non-commonsensical facts which need to be eliminated [Shudo *et al.*, 2016]. This suggests that common sense knowledge should be acquired by interaction, but it needs to be engaging. Last year [Cambria *et al.*, 2016] have proposed GECKA3D game engine for commonsense knowledge acquisition which not only provides a platform for the acquisition of commonsense knowledge, but also enables the development of serious games in 3D. We decided to increase the attractiveness and reality of gamer's movement and created an extendable virtual reality game with purpose of acquiring common sense knowledge.

3 Virtual Reality Game for Knowledge Acquisition

In this section we describe technical details of our prototype. We chose Google Cardboard as the cheapest way to experience VR. It costs only around 20USD and uses a smartphone screen as the display. The device relies on the smartphones gyroscope to detect head movements, and on the compass to detect a magnet button impulse. Our prototype game utilizes this button as the only action related input method. We have tested our game also with Elegiant Headset which is not equipped with a button, so a basic bluetooth input device was used instead (any bluetooth mouse can be used). Smartphone Asus ZenFone2 (Intel 64-bit Super Quad Core) with a relatively large screen (5.5 inches) was chosen to be used as a display with the headsets. We used Google SDK for Android containing the very basics needed to implement VR. Unity 3D[2] was used as the game engine because Google Cardboard provides a SDK for Unity, and it is easy to import any project to Android (or to any VR display). Second author, who is the game developer, chose to write all her scripts in C#. In this stage of development the game was limited to a set of environments for exploration and basic actions. A user can "walk" around these environments and start interacting with items by looking at them. The focus is shown as a reticle which is a dot that becomes a circle when pointed with an action object (see Figure 1).

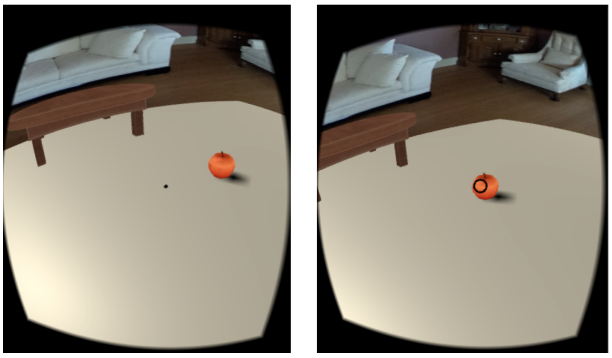


Figure 1: Interaction within the game - reticle pointing on the floor (inert component, on the left), and reticle pointing on an apple (interactive component, on the right).

The player is first presented with a simple Spaceship[3] environment. An animated Alien entity is the only interactive element in this environment, ensuring that the user tries to communicate with it and understands how to interact inside the game. When the host (Alien) is clicked (user points at it with the reticle and presses the magnet button), a dialogue menu is displayed, giving the options to either "Talk" or "Pet" the Alien (the *Pet* function starts an animation of the host activity and it has no impact on the final results; it is meant to engage the player). When *Talk* is chosen, a dialogue menu appears as shown in Figure 2. It introduces the player to the basic game goal: confronted to multiple environments, the player has to check whether some objects belong there or should be moved to another environment. Except the Spaceship area, there are three other environments: Forest, Beach, and Liv-

[1]Onsei Asisuto – Yahoo! Japan. Website. Accessed: 15.5.2017. [Online]. Available: `http://v-assist.yahoo.co.jp`

[2]Unity - Game Engine. Website. Accessed: 15.5.2017. [Online]. Available: `https://unity3d.com/`

[3]We decided to use capital letters for names of environments and objects to distinct them from normal nouns.

Table 1: Environments and the assigned objects.

Forest	Beach	Living Room
Campfire	Umbrella	Stool
Tree	Fish	Book
Apple	Bottle	carpet
Flower	Whale	Piano
Butterfly	Ice Cream	Mug
Mushrooms	Toy Boat	Photo
Folding Chair	Palm Tree	Coffee Table
Iguana	BeachBall	TV-set
Well	Shovel	Sofa
Grass	Rock	Armchair

ing Room (see Figures 3 and 4). They are filled with objects that usually belong (or rarely belong) there and by selecting one or more objects (by clicking it and choosing "select" in the menu) they can be moved to another environment. The objects are moved with the player, thus allowing to put an object in the environment the player thinks it belongs to. When the mission is completed (the player is done moving the objects around to his liking), he can go back to the spaceship and tell the Alien that "Everything is okay" (see Figure 2). This triggers a closing animation and the application shuts down.

Three game environments mentioned above (not including the Spaceship which is used as the main game interface) contain 30 game objects (see Table 1) chosen from ConceptNet5[4] for the terms "forest", "beach" and "living room" (10 objects for each environment, not excluding ones which can belong to more than one context). For example, "AtLocation" and "HasA" relations together with "living room" brought nouns as "TV-set", "sofa", "fireplace" or "table". At the start of each game session, each environment is initialized randomly with a list of 5 objects taken from the 30 game objects available. When the player concludes the game session, a score is attributed to each object and environment. This approach allows us to get a sense of "object belonging to a place" or commonness of a object-place relation, as well as a numerical scale for it. For example, object Folding Chair obtained more points as belonging to the Forest and less for being a part of the Living Room which suggests that while user believes it belongs in the forest, it would not be out of place in the living room. Not only relation scores but players' actions throughout the game session are also saved (what he tried to do with the objects, as they can be taken, dropped, selected or pushed, in which environments he moved and so on). Such session records are stored for future analysis and further extraction of common sense knowledge related to acts and their sequences (scripts).

4 Testing Phase

After several game sessions we analyzed logs and found out that most objects had unanimous scores, categorizing them as belonging to one specific environment only. However,

[4]ConceptNet 5. Website. Accessed: 15.5.2017. [Online]. Available: `conceptnet.io`



Figure 2: Second dialogue menu for the Alien.

amongst the 30 game objects currently available, three had conflicting results:

- The Tree object acquired high score for the Forest but also some points for the Beach.
- The Iguana was also mostly considered as belonging to the Forest (four testers) but two users brought it to the Beach.
- The Folding Chair, as mentioned above, scored higher for belonging to the Forest than to the Living Room environment.

In the case of the Folding Chair, the scores were too close to draw a clear conclusion, and a larger number of tests is needed to determine whether it is a tendency or simply a mistake. The Iguana case was more interesting as the scores were close enough to indicate that people putting an iguana on the beach did not simply made a mistake. Players suggested that while it is most common to see an iguana in the forest, it also would not be considered especially surprising to see one on the beach.

Finally, the case of the Tree differs a little from the previous two. The Tree object used in the game could arguably belong both to the forest or to the beach. Its look is not specific enough to exclude it entirely from the Beach environment or the Forest environment (for reference, another game object is a Palm Tree, which was always moved to the Beach, because player clearly identify it as not belonging to the Forest despite being a kind of tree).

The fact that one of the available environments was a forest seemed to influence players. It could be interesting to



Figure 3: The Forest environment. Randomly distributed objects can be common to the place or very uncommon as the Piano.

remove the Forest environment or to add an option to "trash" game objects if a player decides that they do not belong to any of the available environments. By doing so it could be observed whether a player deem the tree belong to the beach or if it would be better to delete it signalizing that it does not belong to the Beach either. This might help in choosing objects as common to a given environment as possible. Naturally more environments to choose would also be helpful for users and from the knowledge acquisition standpoint. It is worth mentioning that testers were from various cultures - Japanese, German and Polish, although many more experiments would be needed to make statistically significant comparisons. From our observations it is highly probable that there are differences which need wider experiments and further investigation.

5 Problems and Solutions

The game has to be computationally light enough to run smoothly on a smartphone because low frame-rate increases discomfort related to VR as has been confirmed in the past[LaViola, 2000; Kolasinski, 1995]. It can cause so called VR sickness which is a form of motion sickness induced by the dichotomy between what the player sees and the lack of movement his body perceives. If a player is a subject to such problems during her or his first experience with Virtual Reality, it becomes very unlikely that she or he plays again. Sound effects (e.g. footsteps when the player is walking, impact sound when an object is dropped, etc.) help reduce the discomfort and we plan to add them in the near future. Short playing sessions is another way to limit VR sickness – frequent but shorter games present few risks of inducing VR sickness, whereas a continuous session of the same combined length presents high chances of causing this discomfort.

An early phase of prototype testing made apparent the need for detailed instructions in order for the player to act in the expected way. This problem was addressed by adding the Spaceship environment and its Alien which was the interface to give instructions to the player. It seemed logical to us that an extraterrestrial entity would not grasp the appropriate context for our human objects and ask humans for help. We believe it also opens a door for real gamification of the project which in the current state is not satisfactory for most of the players.

One of the problems, especially with the online-based GWAPs, are users who do not follow rules of a game. As a way to prevent malicious players from purposefully placing objects in wrong environments, we plan a data sanity check during the final validation of the results. By establishing a "incorrectness threshold", it should be possible to detect malignant users by comparing results from previous sessions. For example, if more than half object-place assignments differ significantly from the most other users' assignments, the whole sessions results can be deleted or, when scoreboards are added, the mischievous player can be penalized by a significant point deduction. Similar methods are used in crowdsourcing, where e.g. annotators are not paid if their annotations differ too much from the collective average.

6 Possibilities for Story Understanding Task

Everyone's experiences differ, therefore our common sense is not identical; it is a kind of semantic consensus on how things usually happen in a given situation. Textual resources can give us a list of important clues, especially about an abstract phenomena, but usually fails to convey knowledge about common acts details, spatial and physical relations, etc. As image and video analysis are now opening new possibilities for recognizing this kind of knowledge, it is rather difficult to gather world knowledge about interactions or context-embedded discussions. We believe it is probable that because of privacy issues it could be easier, faster and cheaper to equip realistic games (especially these more closely mirroring the real world) with world-learning capabilities, than to make all future robots share their experiences. In our opinion, as sharing knowledge helped human culture to evolve, the same will help artificial intelligence to progress. For both evolutions we need stories which are not in the focus of the current pattern recognition boom in AI. However, available stories are often too laconic for machines to learn from, the miss-

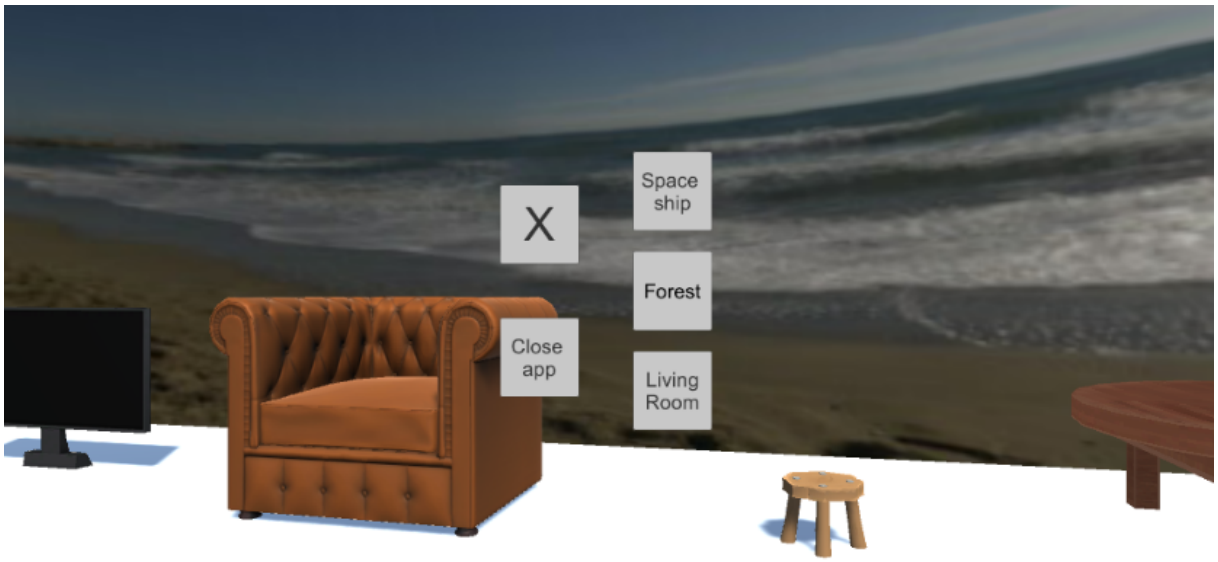


Figure 4: A screenshot of a situation when living room related objects were randomly placed inside the Beach.

ing (unwritten) details should be possible to be discovered in simulations and interactions. Obviously real world simulations may be boring for its users, therefore gamification factor becomes very important. Players could also be given an opportunity to create their own stories, to add objects and their functions, cooperate and exchange. Hopefully virtual and augmented reality will bring a broad range of new ideas for not only collecting data [Dankov *et al.*, 2011] but also using it by artificial agents, e.g. in conversations with users. It is possible that AI projects realized within simulators like Second Life [Goertzel *et al.*, 2007] will be ported to newer platforms and more users will provide valuable feedback. Virtual Reality and Augmented Reality software has already shown its usefulness in teaching people [Bower *et al.*, 2014; Merchant *et al.*, 2014] but surely its potential for teaching machines is in the initial stage and is worth investigating.

7 Conclusions and Future Work

In this paper we described our first stage of prototyping a game for common sense knowledge acquisition. Although we started with a simple object-place matching, our goal is to collect relations between acts and objects, their order and consequences, which is needed for story understanding task. As for the next steps of development and testing our system in the current state, more environments and objects should be added in order to gather broader knowledge of "what is commonly found" in different places. It could also be used to establish "cultural commonsense" by testing the game showing different regions (for example, a typical American living room would be different from a typical Japanese living room).

The game can as well be used to gather commonsense on fictional environments. One could imagine that knowledge about the Harry Potter universe could make an efficient AI for taking care of children, as it would be able to talk about the fantasy world while being inside or outside the game. Players could teach their AI assistants in game simulators and then cooperate with them during the real game and the smarter (well-though) the assistant is, the bigger chances for winning could be given.

Virtual Reality could be particularly useful to gather emotional commonsense knowledge and we plan to extend the task of teaching the Alien not only about objects and places but also to understand human feelings. The first game phase is to present a correctly furnished environment to the player, with a human character in it. That character would display an emotion (for example, a children crying) and the player would choose an object to give to that character in reaction to his emotional state (for example, give an ice-cream or read a book). The second game phase would present to the player the result from someone else's first phase. The player could then choose to, for example, praise or reprimand the character, thus validating or invalidating other players actions (e.g., if the crying children was given a knife, the malicious player should be scolded).

When a more sophisticated game is ready, we plan to release it as a free game to achieve as big number as participant as possible. We believe that giving users a function for adding environments, objects, actors and their actions would be an important first step for collecting world-wide, context-dependent, common sense knowledge. However, to achieve a level of MineCraft sophistication and community size, collaboration with other researchers and companies will be necessary.

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